

Project Name: Legal Ease: AI-Powered Legal Document Generator

Team Members:

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Project Overview & Objective

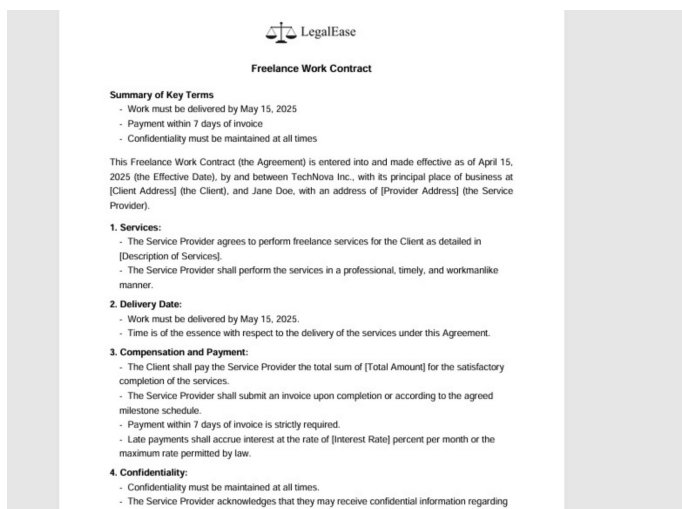
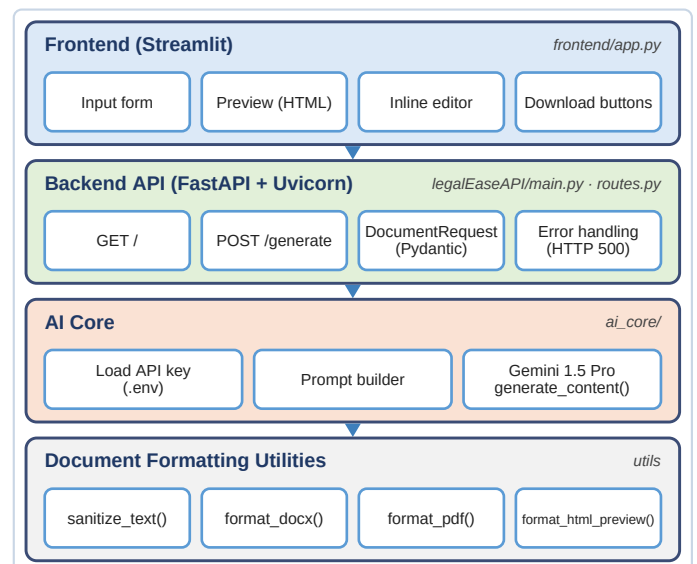
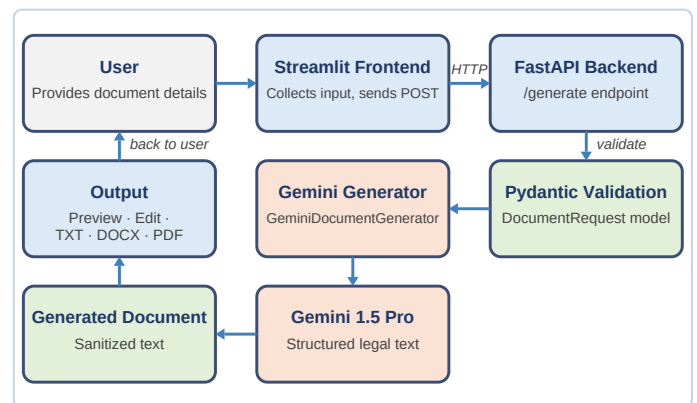
LegalEase is an AI-powered legal document generator developed using Python. It provides a simple web interface where users enter the document type, involved parties, terms and conditions, and effective date to generate a structured legal document.

Objective: Build a simple and useful AI-assisted system that helps users generate structured legal documents from the information they provide.

Main Features

- AI-based legal document generation
- User input for document type, parties, terms and effective date
- FastAPI REST endpoint and Gemini 1.5 Pro integration
- Generated document preview and inline editing
- Download as TXT, DOCX and PDF
- Document formatting and sanitization

Technologies: Python, Streamlit, FastAPI, Gemini 1.5 Pro, Pydantic, python-docx, fpdf2.



Generated legal-document output

Project Architecture & Architecture Description

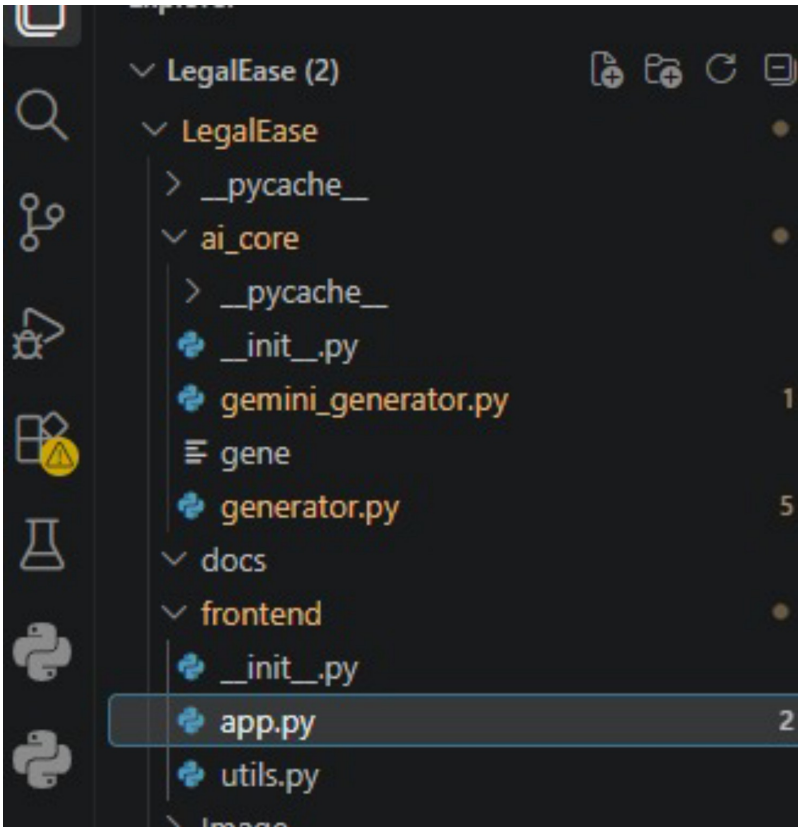
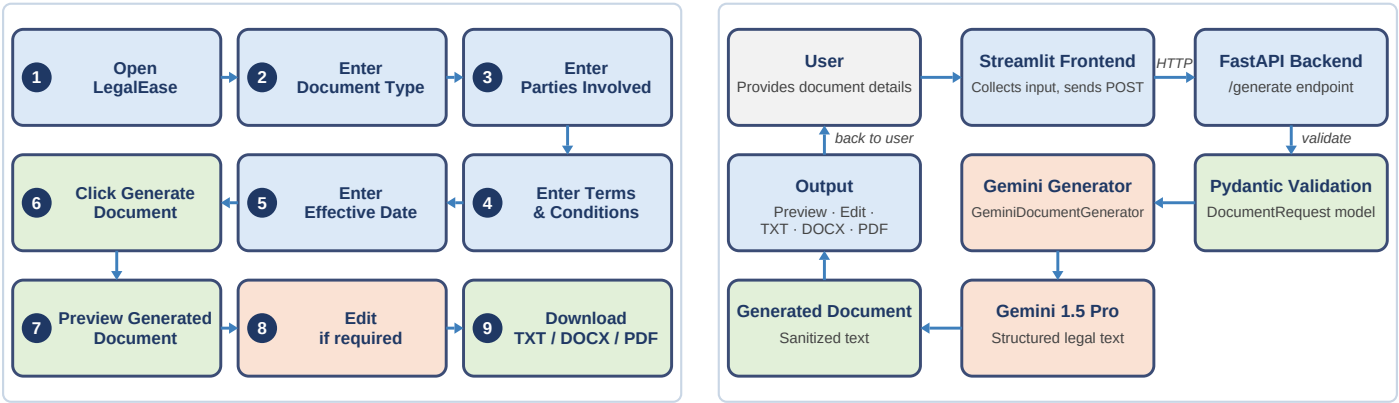
LegalEase follows a modular architecture: the user interface communicates with the backend API, the backend validates the request, and the AI core generates the legal document.

Architecture Flow

USER → STREAMLIT FRONTEND → HTTP POST /generate → FASTAPI BACKEND → GEMINI DOCUMENT GENERATOR → GEMINI 1.5 PRO → GENERATED LEGAL DOCUMENT → PREVIEW / EDIT / TXT / DOCX / PDF

Key Architecture Steps

- User enters the required legal document information.
- Streamlit collects the information and sends it to the backend.
- FastAPI receives the request through the /generate endpoint.
- Pydantic validates the document request data.
- Gemini Document Generator creates a structured prompt.
- Gemini 1.5 Pro generates the legal document text.
- The generated text is sanitized, previewed and edited before download.



Project folder structure

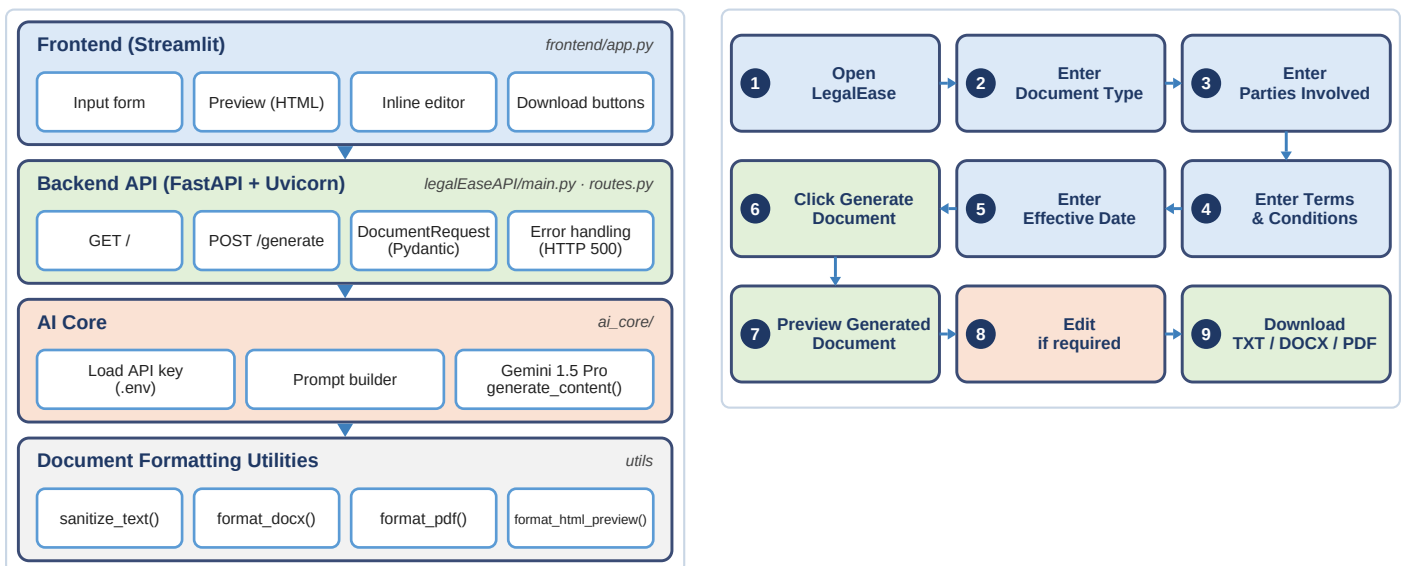
Technical Architecture & System Components

Technical Stack

Python — programming language; Streamlit — frontend framework; FastAPI — backend framework; Uvicorn — API server; Gemini 1.5 Pro — AI model; Pydantic — request validation; python-docx — DOCX creation; fpdf2 — PDF creation; Pillow — image processing; Requests — HTTP communication; python-dotenv — environment configuration.

System Components

- Frontend: collects document type, parties, terms and effective date, then displays the generated document.
- Backend API: provides /generate, receives validated JSON and passes values to the generator.
- AI Core: loads the API key, configures the Google Generative AI service, creates a structured prompt and calls Gemini 1.5 Pro.
- Document Formatting: sanitizes text and prepares DOCX, PDF and HTML preview output.



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LegalEase > requirements.txt
1 fastapi>=0.110
2 uvicorn[standard]>=0.29
3 streamlit>=1.40
4 python-docx>=1.1
5 fpdf2>=2.7.4
6 pillow>=10.0
7 requests>=2.31
8 google-gemai>=1.0
9 python-dotenv>=1.0
10 pydantic>=2.0

(venv) PS C:\Users\acer\Downloads\LegalEase (2)\LegalEase> streamlit run frontend/app.py
2026-09-24 21:23:43.653 Uvicorn server started on ::8503

You can now view your Streamlit app in your browser.

Local URL: http://localhost:8503
Network URL: http://10.195.63.34:8503

Help agents write better Streamlit apps?
Install the official Streamlit skills by running streamlit skills in your terminal.
```

Required Python packages and runtime

Data Flow, Features & Responsibilities

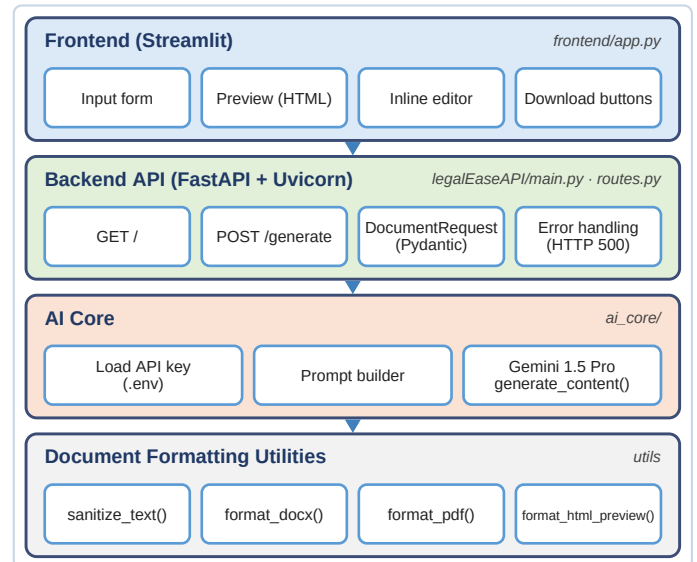
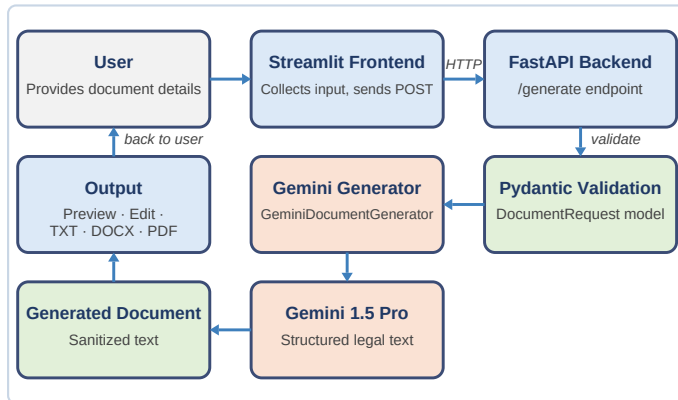
Data Flow: User Input → Streamlit Frontend → HTTP POST Request → FastAPI Backend → Pydantic Validation → Gemini 1.5 Pro → Generated Text → Sanitization → Preview / Edit → File Downloads.

Important Features

- Document type, parties, terms and effective date inputs
- REST API communication
- Gemini 1.5 Pro integration
- Preview and editing
- TXT, DOCX and PDF download
- Error and validation handling

Roles

- User: enters details and generates, edits, previews or downloads the document.
- Streamlit Frontend: collects input, calls the API and provides downloads.
- FastAPI Backend: validates requests and processes /generate.
- Gemini Generator: builds the AI prompt and generates legal text.
- Utility Module: sanitizes and formats TXT/DOCX/PDF/HTML outputs.



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27 class GeminiDocumentGenerator:
28     def generate_document(self, document_type, parties, terms, dates) -> str:
29         message = response.json()["error"]["message"]
30         except Exception:
31             message = response.text[:300]
32             raise GeminiAPIError(f"{response.status_code}: {message}")
33
34         data = response.json()
35         candidates = data.get("candidates") or []
36         if not candidates:
37             raise RuntimeError(f"The model returned no answer: {data.get('promptFeedback')}")
38
39         parts = candidates[0].get("content", {}).get("parts", [])
40         text = "".join(p.get("text", "") for p in parts).strip()
41         if not text:
42             reason = candidates[0].get("finishReason", "unknown")
43             raise RuntimeError(f"The model returned an empty response (reason: {reason})")
44         return text
```

Gemini document-generator implementation

User Flow, Inputs & Project Structure

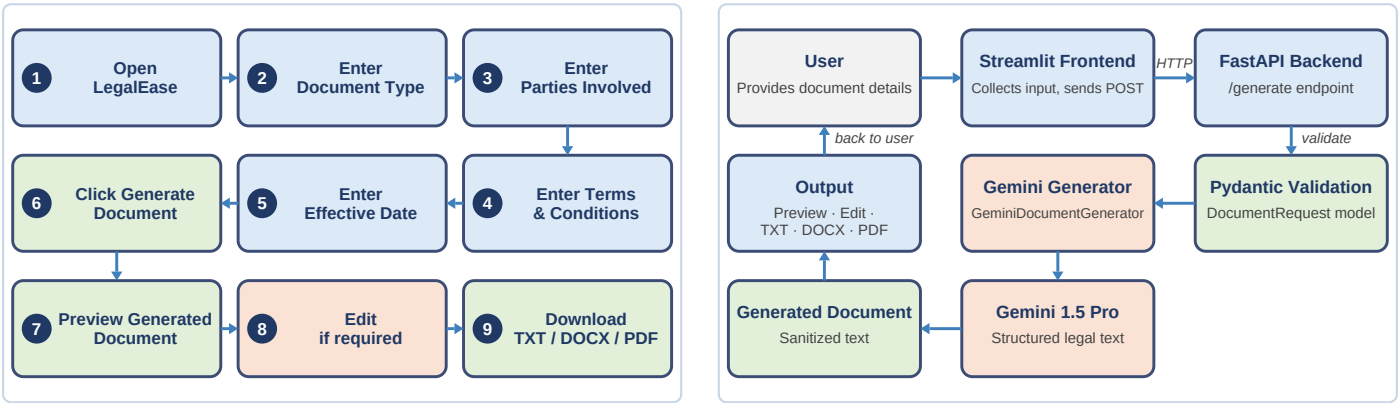
User Flow: Open LegalEase → Enter Document Type → Enter Parties Involved → Enter Terms & Conditions → Enter Effective Date → Click Generate Document → Preview → Edit if required → Download TXT/DOCX/PDF.

Four Main Inputs

- Document Type — specifies the type of legal document.
- Parties — contains the parties involved.
- Terms — contains terms and conditions.
- Dates — provides the effective date.

Folder Structure

- ai_core/ — Gemini-based document generation logic.
- frontend/ — Streamlit interface and formatting utilities.
- legalEaseAPI/ — FastAPI application and API routes.
- Image/ — optional application logo.
- requirements.txt — required Python packages.
- .env.example — environment-variable template.
- README.md — project information and instructions.



AI Legal Document Generator

Document Type (Ex: Agreement, Contract, NDA)

Freelance Work Contract

Parties Involved

Jane Doe (Service Provider), TechNova Inc. (Client)

Terms & Conditions (Use semicolons for bullet points)

Work must be delivered by May 15, 2025; Payment within 7 days of invoice; Confidentiality must be maintained at all times

Streamlit input interface

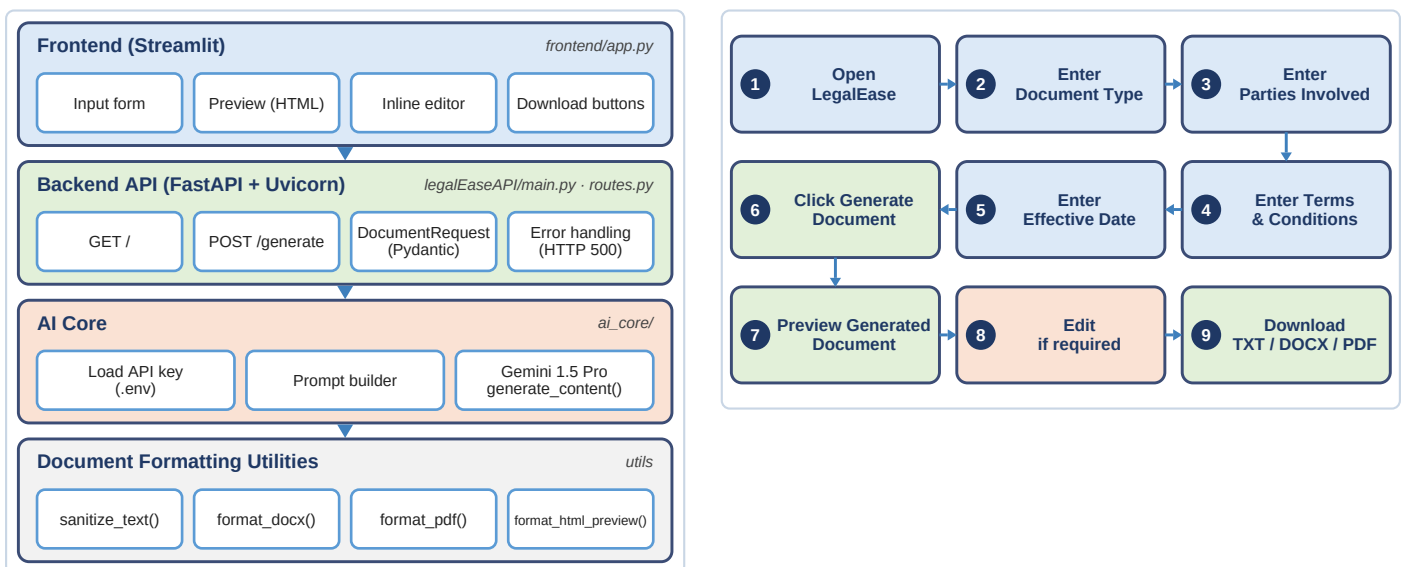
Project Setup & Starting the Application

Setup Steps

- Create virtual environment: `python -m venv venv`
- Activate it on Windows: `venv\Scripts\activate`
- Install dependencies: `pip install -r requirements.txt`
- Rename `.env.example` to `.env` and add `GEMINI_API_KEY`.
- Place a PNG logo at `Image/Logo.png` if required.

Start the Application

- Backend: `uvicorn legalEaseAPI.main:app --reload`
- Frontend: `streamlit run frontend/app.py`
- Documented development address: `http://127.0.0.1:8000`
- Documented local frontend address: `http://localhost:8501`



The screenshot shows the VS Code editor with the project files and the terminal output. The terminal shows the command `streamlit run frontend/app.py` being executed, and the output indicates that the application is running successfully on `http://localhost:8501`.

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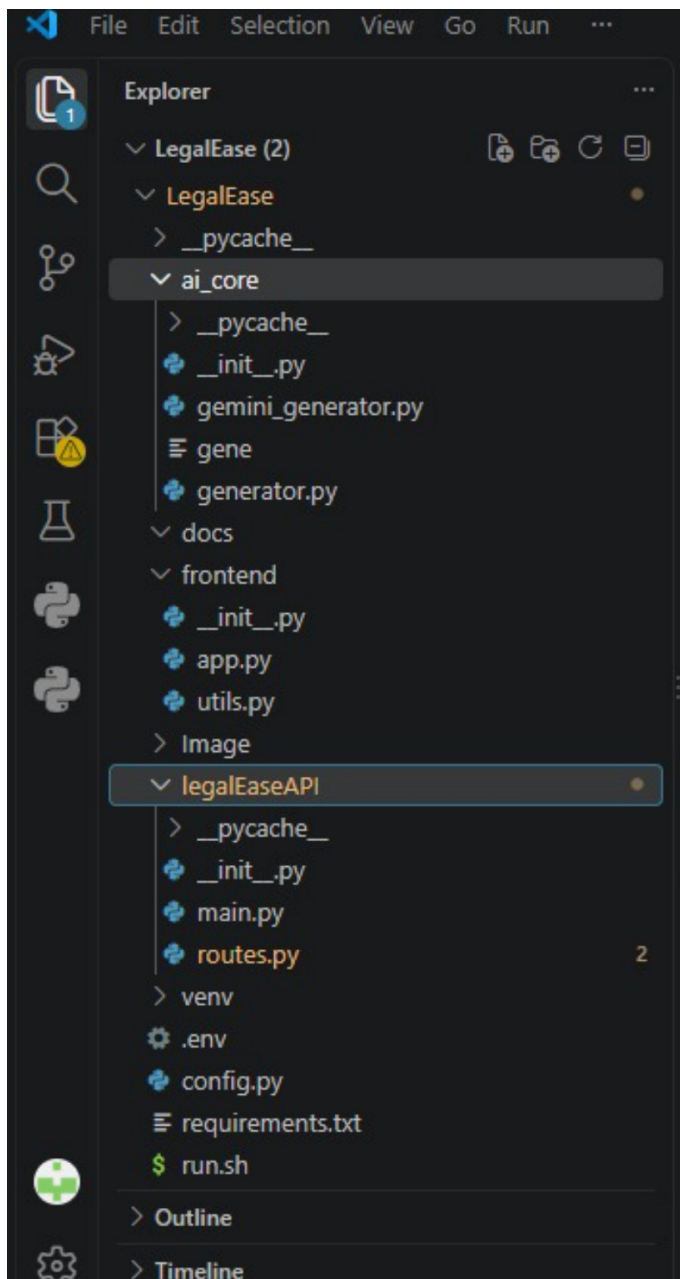
API Endpoints & Backend Development

API Endpoints

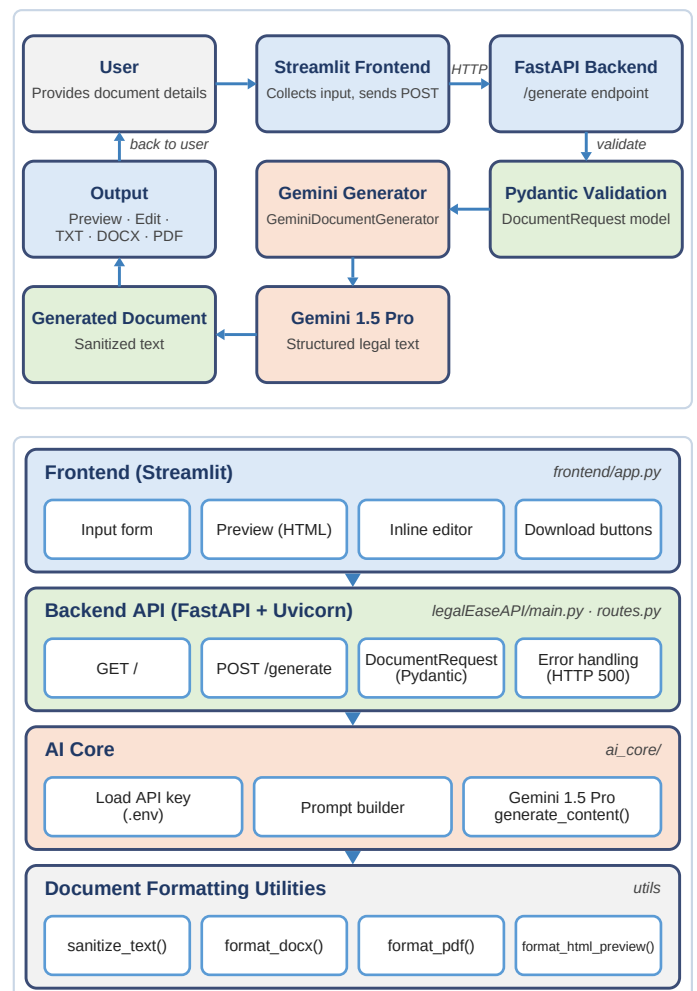
- GET / — API health-check/root endpoint.
- POST /generate — generates a legal document.
- Request fields: document_type, parties, terms, dates.
- The backend returns the generated document in a JSON response using the document field.

Backend Development

- main.py creates the FastAPI application, includes the API router and provides the root health check.
- routes.py defines the DocumentRequest Pydantic model and validates the four input values.
- /generate receives DocumentRequest and calls GeminiDocumentGenerator.generate_document().
- If generation raises an exception, the API returns HTTP 500 with error detail.



FastAPI backend files and project structure



AI, Frontend & Document Formatting

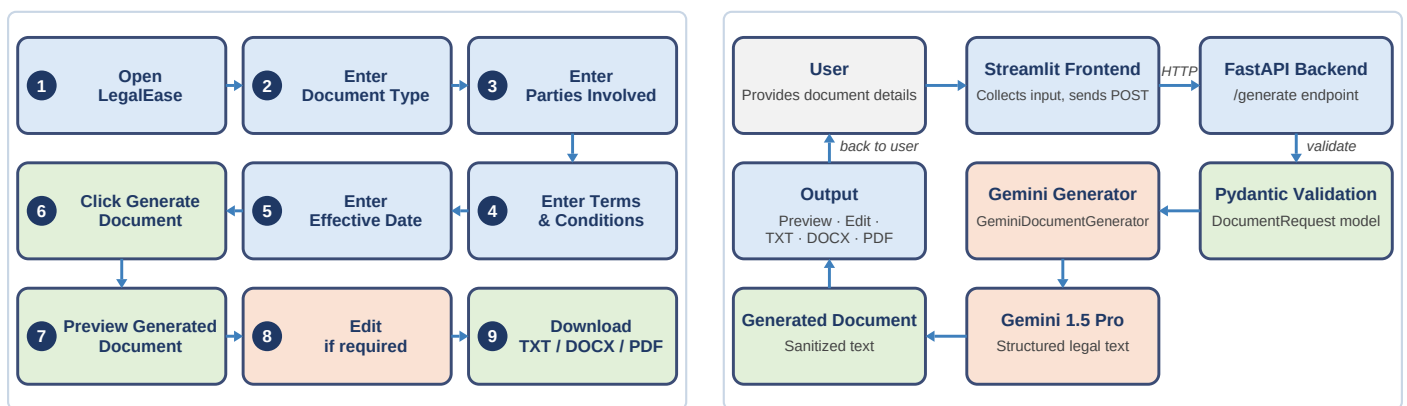
AI Development

- GeminiDocumentGenerator loads environment variables and checks GEMINI_API_KEY.
- It configures the Google Generative AI library and initializes Gemini 1.5 Pro.
- The prompt contains document type, parties, effective date, terms and conditions.
- The generator requests a formal legal structure with sections and clauses.
- The process loads the key, configures the service, builds the prompt, calls generate_content(), and returns the response.

Frontend: app.py creates the LegalEase interface, displays the logo when available, collects inputs, sends the request, stores the generated document in session state, and provides a styled preview/editing area.

Formatting & Export

- Text sanitization replaces incompatible typographic characters.
- DOCX uses a centered title and Times New Roman 11 pt normal text.
- PDF uses fpdf2, centered title, headings and bullet rendering.
- HTML preview converts line breaks for display.
- Download options: TXT, DOCX and PDF.



The screenshot shows a VS Code editor with the following components:

- Explorer:** A file tree on the left showing the project structure, including folders like LegalEase, frontend, and backend, and files like app.py, routes.py, and requirements.txt.
- Editor:** The main workspace showing the implementation of the `sanitize_text` function in `utils.py`. The function takes a string and returns a sanitized version, removing unwanted characters and formatting.
- Terminal:** A terminal window at the bottom showing the output of the application, including the local and network URLs and a message to view the Streamlit app in a browser.

```
1 """
2 frontend/utils.py
3
4 Core formatting utilities used by the Streamlit UI to turn raw
5 AI-generated text into clean, branded, downloadable documents.
6 """
7
8 import io
9 import re
10
11 from docx import Document
12 from docx.shared import Inches, Pt
13 from docx.enum.text import WD_ALIGN_PARAGRAPH
14 from fpdf import FPDF
15
16 from config import WEB_LOGO_PATH
17
18
19 def sanitize_text(text: str) -> str:
```

Document-formatting utility implementation

Execution, Expected Output & Conclusion

Project Execution

- Open the LegalEase project folder in VS Code.
- Create and activate the virtual environment.
- Install dependencies and configure GEMINI_API_KEY.
- Start FastAPI backend and Streamlit frontend.
- Enter document details and generate the legal document.
- Preview/edit the result.
- Download the final document as TXT, DOCX or PDF.

Expected Output: An AI-generated legal document based on the information supplied by the user. The generated content is displayed in the LegalEase interface and can be edited before export.

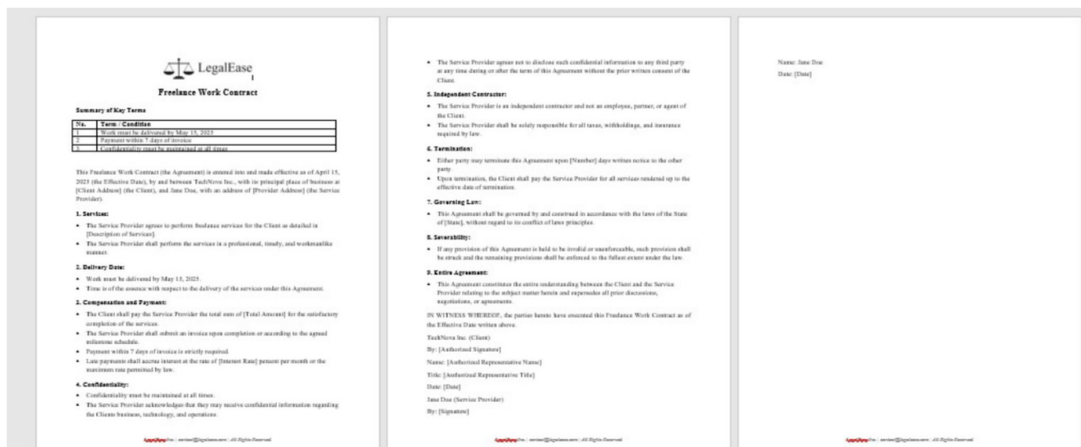
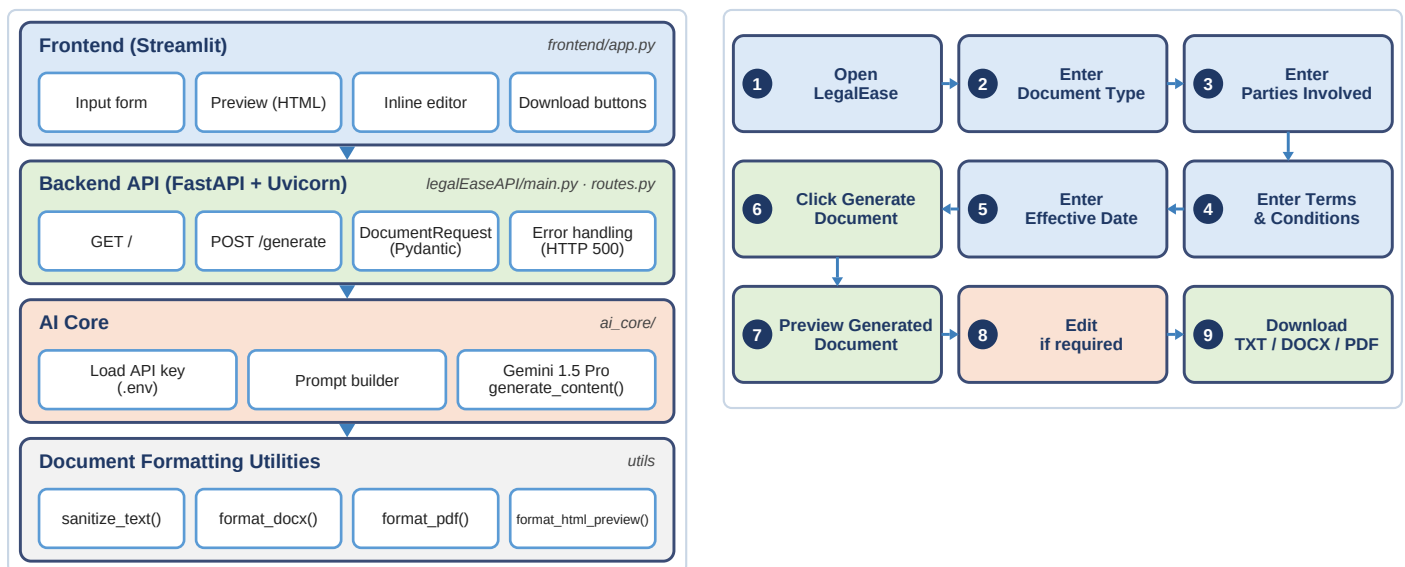
Conclusion

LegalEase combines a Streamlit frontend, FastAPI backend, and Gemini 1.5 Pro to provide an AI-assisted legal document generation workflow. Its modular structure separates the user interface, API layer, AI generation logic, and document-formatting utilities.

Final Workflow: USER INPUT → STREAMLIT → FASTAPI → PYDANTIC VALIDATION → GEMINI 1.5 PRO → GENERATED LEGAL DOCUMENT → PREVIEW / EDIT → TXT / DOCX / PDF

GitHub Link

<https://github.com/bhavadaranim25ds008-collab/LegalEase/tree/main>



Final generated document with signature section